

Dynamic Competition in Elections: A Noncooperative Game of Candidate Positioning

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Abstract—A noncooperative dynamical model of elections is developed. We analyze the case of two candidates competing for a fixed, stationary electorate over a finite horizon. Numerical examples illustrate how voter distribution, campaign duration, and initial positions shape equilibrium candidate trajectories. We extend the model to incorporate credibility losses from policy repositioning.

I. INTRODUCTION

The economic analysis of political behavior traces back to the seminal work of Hotelling [1], which first drew an analogy between spatial competition among firms and the ideological positioning of political candidates. Hotelling’s insight that competitors tend to converge toward the center of the consumer distribution to maximize market share laid the foundation for the spatial theory of voting. Building on this framework, Downs [2] extended the Median Voter Theorem to electoral competition, arguing that in a majoritarian two-party system, rational politicians will converge to the position of the median voter. This spatial model relies critically on the formal results of Black [3], which show that single-peaked preferences yield a stable equilibrium at the median. It is precisely this belief in the existence of a stable equilibrium that underlies the “pendulum democracy” metaphor, which depicts political power as alternating between two competing parties, with the implicit expectation that the amplitude of these oscillations diminishes over time, eventually converging to a stable equilibrium.

The pendulum metaphor is inherently one-dimensional and thus oversimplifies modern electoral settings, where voters evaluate candidates across multiple policy dimensions. While the median voter result yields a robust prediction in unidimensional policy spaces, the existence of equilibrium is significantly challenged in multidimensional settings. In deterministic models, Plott [4] showed that a stable majority-rule equilibrium exists only under highly restrictive symmetry conditions on voter preferences. Moreover, McKelvey [5] established a “chaos theorem,” demonstrating that in the absence of such an equilibrium, any point in the policy space can be reached from any other through a sequence of majority-preferred alternatives. A comprehensive treatment of these spatial constraints and the geometric structure of voting is provided by Enelow and Hinich [6].

To address the instability inherent in deterministic multidimensional models, scholars turned to probabilistic voting

theory. As detailed by Coughlin [7], introducing a stochastic component into voter behavior—arising from imperfect information or non-policy considerations—restores the existence of equilibria even in complex policy spaces under suitable regularity conditions. Lin et al. [8] extended this framework to multicandidate competition, emphasizing how uncertainty shapes candidate positioning. However, the mere existence of equilibrium does not imply convergence to the center. Schofield [9] reconciles the gap between theoretical convergence and observed political divergence by incorporating “valence,” capturing non-policy attributes of candidates, and derives necessary and sufficient conditions for the Mean Voter Theorem, showing that when valence differences across parties are sufficiently large relative to electoral uncertainty, lower-valence candidates are strategically driven away from the electoral center.

Elections, at their core, represent a contest for political office. When there is no obvious best position for candidates to converge to, the system naturally lends itself to analysis through the lens of game theory. There are two timescales in which such games can be considered. *Inter-election dynamics* concern the evolution of the political landscape across successive election cycles — shifts in the electorate, entry and exit of candidates, and the way one election’s outcome shapes the starting conditions of the next. *Intra-election dynamics*, in contrast, concern how candidates reposition themselves within a single campaign: given a fixed electorate and a finite horizon, each candidate adjusts her policy platform over time in response to her opponent’s moves.

In this paper, we focus on the intra-election problem. We model the strategic interactions between candidates as a constant-sum dynamic game, capturing the simultaneous nature of campaign decisions over a finite horizon, and study how candidates optimally navigate the policy space within that period. This model enables a rigorous analysis of campaign strategies in a purely competitive setting, where each candidate’s optimal strategy depends on the anticipated strategies of others.

To the best of our knowledge, existing dynamic models of electoral competition primarily focus on the allocation of campaign resources or on empirical estimation of candidate behavior, while spatial models of policy positioning remain largely static. We bridge these strands by developing a dynamic, noncooperative model of policy positioning in which candidates continuously adjust their platforms over a finite horizon. By characterizing equilibrium candidate trajectories, our approach provides a new perspective on how strategic interactions evolve in multidimensional policy spaces.

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II. MODEL

We model the election campaign as a constant-sum dynamic game. The policy space X is a closed, nonempty, convex subset of $\mathbb{R}^n$ representing the space of policy positions of all candidates and voters' ideal positions. All players, or candidates, observe one another's current positions and simultaneously choose their movements in the policy space, subject to speed constraints, over a campaign with a fixed time horizon T . The players are rational and noncooperative, seeking to maximize their individual expected vote count at the end of the campaign. The terminal payoff of each candidate is her expected vote count, which depends on all players' final positions in the policy space. For the purpose of analysis, we assume that the electorate is finite and has fixed ideal positions during the campaign, and each voter's ideal point is known to all candidates initially. The voters independently make their voting decisions based on the candidates' terminal policy positions. In particular, given the set of players $\mathcal{J} = \{1, \dots, j, \dots, J\}$, an electorate indexed by $\mathcal{I} = \{1, 2, \dots, I\}$, and the players' terminal positions $\mathbf{x}(T) = (x_1(T), \dots, x_J(T)) \in X^J$, the expected vote count V_j of player j is given by the following form,

$$V_j(\mathbf{x}(T)) = \sum_{i \in \mathcal{I}} \Pr[\text{voter } i \text{ votes for player } j \mid \mathbf{x}(T)].$$

We adopt an explicit multinomial logit utility model [9], [10], and discuss variations and potential improvements for the model. The ideal points for the electorate of size I are represented by the set $\{v_i \in X\}_{i \in \mathcal{I}}$ and are observed by all players. Each of the players, or candidates, in the set $\mathcal{J}$, takes a policy position $x_j \in X$. Given a profile of players' positions $\mathbf{x} = (x_1, \dots, x_J) \in X^J$, the utility function of voter i is described by

$$\mathbf{u}_i(v_i, \mathbf{x}) = (u_{i1}(v_i, x_1), \dots, u_{iJ}(v_i, x_J)) \in \mathbb{R}^J,$$

where the utility voter i derives from candidate j is given by

$$u_{ij}(v_i, x_j) = u_{ij}^*(v_i, x_j) + \varepsilon_{ij},$$

$$u_{ij}^*(v_i, x_j) = \lambda_{ij} - \Phi_i(\|v_i - x_j\|_i).$$

u_{ij}^* represents the observable component of voter i 's utility. The term λ_{ij} denotes the exogenously determined valence of candidate j for voter i , while Φ_i is an increasing function from $\mathbb{R}_+$ to $\mathbb{R}$ and $\|\cdot\|_i$ is a norm specific to voter i . The term ε_{ij} represents a stochastic error and we assume that all ε_{ij} are independent and identically distributed according to a Type I extreme value distribution Ψ , with the cumulative distribution function $F_\Psi(h) = \exp(-\exp(-h))$ for $h \in \mathbb{R}$.

Given this stochastic uncertainty, the probability that voter i with ideal position v_i votes for candidate j is:

$$s_{ij} = \Pr[u_{ij}(v_i, x_j) > u_{il}(v_i, x_l), l \neq j].$$

The expected vote count of player j is $V_j(\mathbf{x}) = \sum_{i \in \mathcal{I}} s_{ij}$, where $\mathbf{x} = (x_1, \dots, x_J)$ collects the positions of all players.

In the model given above, the expected vote count for player j is,

$$V_j(\mathbf{x}) = \sum_{i=1}^I \frac{\exp(u_{ij}^*(v_i, x_j))}{\sum_{k=1}^J \exp(u_{ik}^*(v_i, x_k))}.$$

The function $V : X^J \rightarrow \mathbb{R}^J, \mathbf{x} \mapsto (V_1(\mathbf{x}), \dots, V_J(\mathbf{x}))$ is a payoff mapping. We consider the dynamic setting with the above payoff function as the terminal payoff. The campaign for the election has a fixed time horizon T . Each candidate j dynamically changes her policy position $x_j(t)$ in space X during the campaign, while observing the other candidates' current positions. To reflect the practical constraint that policy repositioning cannot be instantaneous, each candidate is subject to a speed constraint c_j . There is no stage cost, and the terminal utility functions are given by the payoff function $V(\mathbf{x})$.

Remark 2.1 (Static pure strategy Nash equilibrium): If X is nonempty, compact, and convex, and every V_j is continuous in the full strategy profile and quasiconcave in x_j with fixed x_{-j} , the Debreu–Glicksberg–Fan theorem [11] guarantees a pure Nash equilibrium. The probability of a single voter s_{ij} is quasiconcave with respect to x_j with a convex increasing function Φ_i , while fixing other players' positions. However, the summation of quasiconcave functions is not necessarily quasiconcave. As a result, we cannot conclude the existence of a pure strategy Nash equilibrium of the model in the static setting.

Remark 2.2 (Vote for the closest candidate): We assume that λ_{ij} is constant and take $\Phi_i : y \mapsto Cy$. As $C \rightarrow \infty$, a uniquely closest candidate is selected with probability tending to one. When several candidates are tied at the minimum distance and have equal valence, they share the limiting choice probability equally.

For simplification, we only consider the case where there are two players, P_1 and P_2 . As a result, the utility functions are given by the expected vote count; we formulate the election as a two-person constant-sum dynamic game with a maximizer and a minimizer player. We will theoretically detail the two-person constant-sum dynamic game in the remainder of the paper.

We consider the continuous-time setting, and will discuss discretized numerical examples in Section IV. The election duration is fixed at T , and we take the state space $X^2 = \mathbb{R}^{2n}$ for the simplification of analysis, stacking the two candidates' positions (the first n entries for candidate P_1 and the last n for P_2). The strategy sets are given by, for $j = 1, 2$,

$$\Gamma_j = \{\gamma : [0, T] \times \mathbb{R}^{2n} \rightarrow \mathbb{R}^n : \gamma \text{ is } C^1 \text{ and } \|\gamma(t, \mathbf{x})\|_s \leq c_j, \forall t \in [0, T], \mathbf{x} \in \mathbb{R}^{2n}\}.$$

Each candidate j chooses a continuously differentiable (pure) feedback strategy with a speed constraint on a norm $\|\cdot\|_s$ from the feasible set Γ_j . For any given pair of pure strategies (γ_1, γ_2) , the state evolves according to,

$$\begin{aligned} \dot{\mathbf{x}}(t) &= f(t, \mathbf{x}(t), a_1(t), a_2(t)) = (a_1(t), a_2(t)), \\ a_j(t) &= \gamma_j(t, \mathbf{x}(t)), \quad j = 1, 2. \end{aligned}$$

Here, f is the general dynamics notation used in dynamic games, and $a_j(t) \in \mathbb{R}^n$ is the action of player j at time t , induced by the feedback strategy γ_j at the current state $\mathbf{x}(t)$. We denote the objective functional,

$$L(\gamma_1, \gamma_2) = \int_0^T g(t, \mathbf{x}(t), a_1(t), a_2(t)) dt + q(\mathbf{x}(T)).$$

In this dynamic game, we model two players, P_1 and P_2 , as rational agents seeking to maximize their individual benefit. With a fixed electorate, the total expected vote count is constant, and for the baseline model we take $g \equiv 0$ (no stage cost); the model then becomes a two-person, constant-sum dynamic game. Let q be the expected vote count (depending only on $\mathbf{x}(T) = (x_1(T), x_2(T))$) for candidate P_2 . Consequently, P_2 acts as the maximizer, aiming to increase q , while P_1 acts as the minimizer. The objective functional L , representing P_2 's expected vote count based on the players' final positions $(x_1(T), x_2(T))$, takes the form:

$$L = q(\mathbf{x}(T)) = q(x_1(T), x_2(T)), \\ = \sum_{i \in \mathcal{I}} \frac{\exp(u_{i2}^*(v_i, x_2(T)))}{\exp(u_{i1}^*(v_i, x_1(T))) + \exp(u_{i2}^*(v_i, x_2(T)))}.$$

A. Threshold model

The multinomial logit model with continuous Φ_i allows for continuous substitution between candidates: every voter assigns some positive probability to each candidate, no matter how far away. A natural question is what happens when voters have sharp preferences—they either find a candidate acceptable or they do not.

To formalize this, we take a limiting case of the logit model for the two candidates, P_1 (minimizer) and P_2 (maximizer). Throughout this subsection, all voters share a common norm $\|\cdot\|_p$ on $\mathbb{R}^n$, i.e., $\|\cdot\|_i \equiv \|\cdot\|_p$. Set the valences λ_{ij} to be constant across candidates, and let $r(v) > 0$ be the acceptance threshold shared by all voters with ideal position v . Define

$$\Phi_{M,v}(d) = M \cdot \mathbf{1}\{d > r(v)\}$$

for a sharpness parameter $M > 0$, and take $\Phi_i = \Phi_{M,v_i}$. As $M \rightarrow \infty$, the logit choice probability s_{i2} converges to

$$s_{i2} \rightarrow \begin{cases} 1, & \text{if } \|v_i - x_2\|_p \leq r(v_i), \|v_i - x_1\|_p > r(v_i); \\ 0, & \text{if } \|v_i - x_1\|_p \leq r(v_i), \|v_i - x_2\|_p > r(v_i); \\ \frac{1}{2}, & \text{otherwise.} \end{cases}$$

This defines a *threshold utility model*: a voter with ideal position v finds candidate j acceptable if and only if $\|x_j - v\|_p \leq r(v)$. A voter who finds exactly one candidate acceptable votes deterministically for that candidate; a voter who finds both or neither acceptable is indifferent and splits evenly between the two.

Let $\mathbf{P}$ be the population measure of voters over ideal positions in X . The finite electorate of Section II corresponds to $\mathbf{P} = \sum_{i \in \mathcal{I}} \delta_{v_i}$. To cast the constant-sum game as a zero-sum game, we shift the payoff so that indifferent voters

contribute zero. The net payoff from voters at ideal position v , given terminal positions $\mathbf{x} = (x_1, x_2)$, is

$$w_v(x_1, x_2) = \begin{cases} 1/2, & \text{if } \|x_2 - v\|_p \leq r(v) \text{ and } \|x_1 - v\|_p > r(v), \\ -1/2, & \text{if } \|x_2 - v\|_p > r(v) \text{ and } \|x_1 - v\|_p \leq r(v), \\ 0, & \text{otherwise.} \end{cases}$$

The terminal payoff is given by,

$$q(x_1, x_2) = \int_X w_v(x_1, x_2) \mathbf{P}(dv).$$

The sharp threshold structure has an important analytical consequence. Within the acceptance region $\|v - x_j\|_p \leq r(v)$, utility is flat: a voter who finds both candidates acceptable is indifferent regardless of their exact distances. This eliminates the cross-dependence between x_1 and x_2 in the payoff and indicates a property that, as we show in Section III, leads to a complete decoupling of the two candidates' saddle-point strategies. By contrast, a Φ such as the capped quadratic form used in Section IV for numerical computation retains gradient information within the acceptance region, coupling the players' strategies through the softmax denominator and preventing decoupling in general.

III. ANALYSIS

In this section, we analyze the multinomial logit model and the threshold model introduced in Section II.

A. Multinomial logit model

For the purpose of analysis, we take the policy space $X = \mathbb{R}^n$. Recall that $\|\cdot\|_s$ denotes the norm on $\mathbb{R}^n$ defining the speed constraints in Γ_1 and Γ_2 , and write

$$\|\omega\|_{s,*} := \max_{\|a\|_s \leq 1} \langle \omega, a \rangle, \quad \omega \in \mathbb{R}^n,$$

for its dual norm, where $\langle \cdot, \cdot \rangle$ is the Euclidean inner product. The unit ball $\{a : \|a\|_s \leq 1\}$ is compact and symmetric, so for every $c > 0$ and every $\omega \in \mathbb{R}^n$ the extrema

$$\min_{\|a\|_s \leq c} \langle \omega, a \rangle = -c \|\omega\|_{s,*}, \quad \max_{\|a\|_s \leq c} \langle \omega, a \rangle = c \|\omega\|_{s,*} \quad (1)$$

are attained.

Let $W : [0, T] \times \mathbb{R}^{2n} \rightarrow \mathbb{R}$ denote a candidate value function of the game, and let $\partial W / \partial x_j$ denote the partial gradient of W with respect to x_j (the j -th block of n entries). The upper and lower Bellman–Isaacs equations associated with the game are

$$-\frac{\partial W}{\partial t} = \min_{\|a_1\|_s \leq c_1} \max_{\|a_2\|_s \leq c_2} \left\langle \frac{\partial W}{\partial \mathbf{x}}, (a_1, a_2) \right\rangle, \quad (2)$$

$$= \max_{\|a_2\|_s \leq c_2} \min_{\|a_1\|_s \leq c_1} \left\langle \frac{\partial W}{\partial \mathbf{x}}, (a_1, a_2) \right\rangle, \quad (3)$$

for all $t \in [0, T)$ and $\mathbf{x} \in \mathbb{R}^{2n}$, with terminal condition $W(T, \mathbf{x}) = q(\mathbf{x})$ for all $\mathbf{x}$. The Hamiltonian is additively separable; hence the lower and upper Hamiltonians coincide. We have

$$\left\langle \frac{\partial W}{\partial \mathbf{x}}, (a_1, a_2) \right\rangle = \left\langle \frac{\partial W}{\partial x_1}, a_1 \right\rangle + \left\langle \frac{\partial W}{\partial x_2}, a_2 \right\rangle,$$

and the two action sets are independent, so the minimum and the maximum act on disjoint blocks of variables and interchange, which implies that (2) and (3) coincide and hence the Isaacs condition holds. Evaluating the two extrema by (1) reduces (2)–(3) to the Hamilton–Jacobi–Isaacs (HJI) equation (4) below. The following dynamic game verification result, in the spirit of [12, Cor. 6.6], gives a sufficient condition for a feedback saddle point, conditional on a solution of that equation.

Proposition 3.1 (Verification theorem): Assume:

- (i) there exists $W \in C^1([0, T] \times \mathbb{R}^{2n})$ satisfying the HJI equation

$$-\frac{\partial W}{\partial t} = -c_1 \left\| \frac{\partial W}{\partial x_1} \right\|_{s,*} + c_2 \left\| \frac{\partial W}{\partial x_2} \right\|_{s,*} \quad (4)$$

for all $(t, \mathbf{x}) \in [0, T] \times \mathbb{R}^{2n}$, with terminal condition $W(T, \cdot) = q$;

- (ii) there exist admissible strategies $\gamma_1^* \in \Gamma_1$, $\gamma_2^* \in \Gamma_2$ attaining the extrema in (1) pointwise, i.e., for all $(t, \mathbf{x})$,

$$\begin{aligned} \left\langle \frac{\partial W}{\partial x_1}, \gamma_1^*(t, \mathbf{x}) \right\rangle &= -c_1 \left\| \frac{\partial W}{\partial x_1} \right\|_{s,*}, \\ \left\langle \frac{\partial W}{\partial x_2}, \gamma_2^*(t, \mathbf{x}) \right\rangle &= c_2 \left\| \frac{\partial W}{\partial x_2} \right\|_{s,*}. \end{aligned}$$

Then (γ_1^*, γ_2^*) is a feedback saddle point: for every initial state $\mathbf{x}(0)$ and all $\gamma_1 \in \Gamma_1$, $\gamma_2 \in \Gamma_2$,

$$L(\gamma_1^*, \gamma_2) \leq L(\gamma_1^*, \gamma_2^*) \leq L(\gamma_1, \gamma_2^*),$$

where $L(\gamma_1, \gamma_2) = q(\mathbf{x}(T))$ under the closed-loop dynamics from $\mathbf{x}(0)$. In particular, the game has the value $L(\gamma_1^*, \gamma_2^*) = W(0, \mathbf{x}(0))$.

Hypothesis (ii) asks for an admissible selection of the extremizers in (1). In particular, when the speed norm is Euclidean, $\|\cdot\|_s = \|\cdot\|_2$, it is self-dual, and wherever the partial gradients are nonzero, the gradient-following pointwise extremizers are unique

$$\gamma_1^* = -c_1 \frac{\partial W / \partial x_1}{\|\partial W / \partial x_1\|_2}, \quad \gamma_2^* = c_2 \frac{\partial W / \partial x_2}{\|\partial W / \partial x_2\|_2}.$$

The norm $\|\cdot\|_s = \|\cdot\|_\infty$ has dual $\|\cdot\|_1$, and the extremizers are componentwise bang–bang,

$$(\gamma_1^*)_k = -c_1 \operatorname{sgn} \left(\frac{\partial W}{\partial x_1} \right)_k, \quad (\gamma_2^*)_k = c_2 \operatorname{sgn} \left(\frac{\partial W}{\partial x_2} \right)_k,$$

for every component k with $(\partial W / \partial x_j)_k \neq 0$. These formulas need not be C^1 at zero-gradient surfaces. Proposition 3.1 yields a feedback saddle point only when the pointwise formulas admit selections in Γ_1 and Γ_2 .

Proof: Write $p_j = \partial W / \partial x_j$. By (1), hypothesis (i) states that

$$\frac{\partial W}{\partial t} + \min_{\|a_1\|_s \leq c_1} \langle p_1, a_1 \rangle + \max_{\|a_2\|_s \leq c_2} \langle p_2, a_2 \rangle = 0,$$

and hypothesis (ii) states that γ_1^* and γ_2^* attain the minimum and the maximum pointwise.

Fix the initial state $\mathbf{x}(0)$ and let $\gamma_2 \in \Gamma_2$ be arbitrary. Because γ_1^* and γ_2 are C^1 , the closed-loop vector field is continuous and locally Lipschitz in $\mathbf{x}$, so by Picard’s

existence theorem the initial value problem has a unique maximal solution. The speed bounds make $\dot{\mathbf{x}}$ uniformly bounded, so the solution remains in a compact set and, by the continuation theorem, extends to all of $[0, T]$. As $W \in C^1$, the map $t \mapsto W(t, \mathbf{x}(t))$ is C^1 with

$$\begin{aligned} \frac{d}{dt} W(t, \mathbf{x}(t)) &= \frac{\partial W}{\partial t} + \langle p_1, \gamma_1^*(t, \mathbf{x}(t)) \rangle \\ &\quad + \langle p_2, \gamma_2(t, \mathbf{x}(t)) \rangle \\ &\leq \frac{\partial W}{\partial t} - c_1 \|p_1\|_{s,*} + c_2 \|p_2\|_{s,*} = 0, \end{aligned}$$

where the inequality uses hypothesis (ii) for the γ_1^* term (with equality) and (1) for the γ_2 term (since $\|\gamma_2(t, \mathbf{x})\|_s \leq c_2$). Integrating over $[0, T]$ and using $W(T, \cdot) = q$,

$$L(\gamma_1^*, \gamma_2) = q(\mathbf{x}(T)) = W(T, \mathbf{x}(T)) \leq W(0, \mathbf{x}(0)).$$

The symmetric argument under (γ_1, γ_2^*) for arbitrary $\gamma_1 \in \Gamma_1$ gives $L(\gamma_1, \gamma_2^*) \geq W(0, \mathbf{x}(0))$. Finally, under (γ_1^*, γ_2^*) both terms attain their extrema, so $\frac{d}{dt} W(t, \mathbf{x}(t)) \equiv 0$ and $L(\gamma_1^*, \gamma_2^*) = W(0, \mathbf{x}(0))$. Combining the three relations yields the saddle-point inequalities and the value of the game. ■

These are sufficient verification conditions. The electoral primitives do not guarantee a classical solution or a smooth admissible selection of extremizers in general; the finite games in Section IV are solved separately by backward induction.

B. Threshold model

Recall that we use $\|\cdot\|_s$ and $\|\cdot\|_p$ to represent the norm used in the speed constraint for candidates and voters’ threshold, respectively.

Theorem 3.2 (Decoupling of saddle-point strategies):

Consider the two-player zero-sum dynamic game with the threshold utility model defined in Section II-A, under the regularity assumptions that $r : X \rightarrow \mathbb{R}_+$ is measurable, and the population measure $\mathbf{P}$ is a finite Borel measure on X . The terminal payoff decomposes as

$$\begin{aligned} q(x_1(T), x_2(T)) &= \frac{1}{2} \mathbf{P}(\{v : \|x_2(T) - v\|_p \leq r(v)\}) \\ &\quad - \frac{1}{2} \mathbf{P}(\{v : \|x_1(T) - v\|_p \leq r(v)\}). \end{aligned}$$

Consequently, the saddle-point optimization decouples. For each initial state, a coverage-maximizing terminal position together with the corresponding constant control yields an opponent-independent saddle point:

$$x_j^* \in \operatorname{argmax}_{y \in R_j} \mathbf{P}(\{v : \|y - v\|_p \leq r(v)\}),$$

where $R_j = \{y \in X : \|y - x_j(0)\|_s \leq Tc_j\}$ is the reachable set of candidate j , independently of the opponent’s initial position $x_{-j}(0)$ and strategy γ_{-j} . In particular, the value of the game is

$$\frac{1}{2} \mathbf{P}(\{v : \|x_2^* - v\|_p \leq r(v)\}) - \frac{1}{2} \mathbf{P}(\{v : \|x_1^* - v\|_p \leq r(v)\}).$$

Proof: Recall that this is a zero-sum dynamic game in which P_1 is the minimizer with final position $x_1(T)$ and P_2 is the maximizer with final position $x_2(T)$. For every v ,

$$w_v(x_1(T), x_2(T)) = \frac{1}{2} \mathbf{1}\{\|x_2(T) - v\|_p \leq r(v)\} - \frac{1}{2} \mathbf{1}\{\|x_1(T) - v\|_p \leq r(v)\},$$

since the two indicators coincide exactly when v finds both candidates acceptable or neither. Integrating over X ,

$$q(x_1(T), x_2(T)) = \frac{1}{2} \mathbf{P}(\{v : \|x_2(T) - v\|_p \leq r(v)\}) - \frac{1}{2} \mathbf{P}(\{v : \|x_1(T) - v\|_p \leq r(v)\}).$$

The set of terminal positions reachable by candidate j over the horizon $[0, T]$ is exactly R_j : every $y \in R_j$ is reached by the constant strategy $\gamma_j(t, x) \equiv (y - x_j(0))/T$, which is admissible in Γ_j , and no admissible strategy moves candidate j farther than Tc_j from $x_j(0)$ in the speed norm $\|\cdot\|_s$. Hence, the value of the game is given by,

$$\begin{aligned} & \min_{y_1 \in R_1} \max_{y_2 \in R_2} \frac{1}{2} \mathbf{P}(\{v : \|y_2 - v\|_p \leq r(v)\}) \\ & \quad - \frac{1}{2} \mathbf{P}(\{v : \|y_1 - v\|_p \leq r(v)\}) \\ & = \max_{y_2 \in R_2} \frac{1}{2} \mathbf{P}(\{v : \|y_2 - v\|_p \leq r(v)\}) \\ & \quad - \max_{y_1 \in R_1} \frac{1}{2} \mathbf{P}(\{v : \|y_1 - v\|_p \leq r(v)\}). \end{aligned}$$

The maxima are attained: for each fixed v the acceptance set $\{y : \|y - v\|_p \leq r(v)\}$ is closed, so the coverage map $y \mapsto \mathbf{P}(\{v : \|y - v\|_p \leq r(v)\})$ is upper semicontinuous since $\mathbf{P}$ is finite, and each R_j is compact. Candidate j , with speed limit c_j , horizon T , and initial position $x_j(0)$, therefore moves to

$$\begin{aligned} x_1^* & \in \operatorname{argmax}_{y \in R_1} \frac{1}{2} \mathbf{P}(\{v : \|y - v\|_p \leq r(v)\}), \\ x_2^* & \in \operatorname{argmax}_{y \in R_2} \frac{1}{2} \mathbf{P}(\{v : \|y - v\|_p \leq r(v)\}). \end{aligned}$$

Because the terminal payoff decomposes into a term depending only on $x_1(T)$ and a term depending only on $x_2(T)$, the minimax and maximin problems separate. Each candidate's saddle-point strategy reduces to maximizing her own voter coverage over her reachable set, independently of the opponent. This completes the proof. ■

IV. NUMERICAL EXAMPLES

In this section, we provide numerical examples to demonstrate insights into our dynamic game model under different settings. The primary insight from our computational experiments in this section is that candidate behavior is shaped by the interaction of three factors: the spatial distribution of voters, the campaign horizon T , and the candidates' initial positions. These factors jointly determine whether candidates converge toward high-density voter regions or compete over local clusters.

TABLE I

PARAMETERS OF THE NUMERICAL EXAMPLES IN SECTION IV. IN THE UTILITY MODEL, v_i IS THE IDEAL POINT OF VOTER i , x_j IS THE POSITION OF CANDIDATE j . THE VALENCES ARE SET TO $\lambda_{ij} = 0$. CANDIDATE-SPECIFIC SPATIAL UTILITY IS CAPPED BEYOND $r_i = r(v_i)$; WHEN BOTH CANDIDATES ARE BEYOND r_i , THEIR SPATIAL UTILITIES COINCIDE BECAUSE VALENCE IS ZERO.

Grid size	Utility model $u_{ij}^*(v_i, x_j)$	No. of voters	Space
$G = 21$	$-60(\min\{\ v_i - x_j\ _2, r_i\})^2$	70	$[0, 1]^2$

For computational purposes, we consider a discrete-time, discretized-state version of the continuous-time dynamic game described in Section II. We model the policy space as the unit square for practical computation. Each stage has unit length ($\Delta t = 1$), so the horizon T coincides with the number of stages, and the policy space is discretized into a grid with G points in each dimension. At each stage, candidates may move to any neighboring grid point or remain stationary. The speed constraint is enforced by restricting movement to these adjacent points (including diagonal moves), yielding up to 3^n actions for each candidate at each stage, where n is the dimension of the policy space. The objective function remains the expected vote count at the terminal stage, as discussed in Section II. The specific parameters used in our numerical examples are summarized in Table I. Given the utility model u_{ij}^* and the players' terminal positions $x_1(T), x_2(T)$, the expected vote count for candidate P_2 is computed as

$$V_2(x_1(T), x_2(T)) = \sum_{i \in \mathcal{I}} \frac{\exp(u_{i2}^*(v_i, x_2(T)))}{\sum_{k=1}^2 \exp(u_{ik}^*(v_i, x_k(T)))}.$$

Since each candidate possesses a finite action set, we can compute the mixed-strategy Nash equilibrium at each time-state pair, provided the value function for the subsequent stage is known. We compute the value function for all stages via backward induction. Finally, to simulate the candidates' trajectories forward in time, we sample actions according to the finite zero-sum stage-game saddle point derived at each stage.

Relation to the continuous-time game. The computations return to the coupled multinomial-logit game of Section II, rather than the decoupled threshold limit of Theorem 3.2. Let $\mathcal{X}_h$ denote the policy grid with mesh h and let $N\Delta t = T$. Starting from $W_N^h = q$, backward induction defines W_k^h at each joint state as the mixed value of the zero-sum matrix whose (δ_1, δ_2) entry is $W_{k+1}^h(x_1 + \delta_1, x_2 + \delta_2)$, where $\delta_j \in h\{-1, 0, 1\}^n$ is a feasible grid displacement. This recursion is the discrete Bellman–Isaacs counterpart of (4): it preserves the state variables, simultaneous feedback interaction, campaign horizon, and terminal logit payoff. The reported runs use $h = 0.05$ and $\Delta t = 1$, so $c_j = 0.05$.

A. Voter distribution

Comparing Fig. 1(a) with Fig. 1(b), where the horizon and initial positions are held fixed but the electorate changes in both its ideal-point distribution and its radii, we observe that candidates shift from converging toward the center to

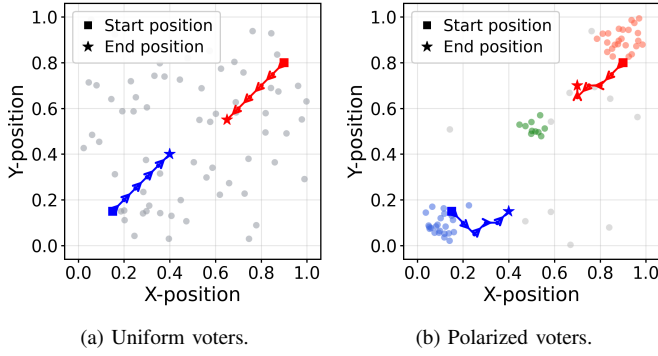


Fig. 1. Effect of voter distribution on candidate trajectories with horizon $T = 5$. The stagewise mixed-strategy Nash equilibria along the displayed paths are pure. The candidates' initial positions are $x_1(0) = (0.15, 0.15)$, $x_2(0) = (0.90, 0.80)$. (a) Under uniformly distributed voters on $[0, 1]^2$ with threshold $r = 0.5$, both candidates move toward the center. Dots: voter ideal points; blue and red paths: candidate positions from $t = 0$ to $t = T$. (b) Under a polarized electorate, the candidates remain separated, each remaining near her initial cluster while contesting the background voters. Blue and red dots: polarized spatial voters ($r = 1.2$); green dots: centrist voters ($r = 0.1$); grey dots: moderate voters ($r = 0.5$).

competing over polarized spatial clusters. In the uniform electorate case, both selected paths move toward the central region, resembling the centripetal tendency familiar from one-dimensional median-voter models. Polarization breaks this logic: the payoff landscape fragments into local peaks, and candidates are drawn toward whichever cluster they can most effectively contest given their current position. Polarization thus transforms the game from a single tug-of-war over the center into a collection of localized contests, where each candidate must decide which cluster to prioritize—a decision that, as we show next, depends critically on both the time horizon and the starting position.

B. Campaign horizon

Comparing Fig. 1(b) with Fig. 2(a), where only the horizon increases from $T = 5$ to $T = 7$, both candidates pivot from polarized spatial clusters toward centrist voters—a discrete shift in equilibrium behavior driven by a change in the time budget. The finite horizon introduces a reachability constraint that restricts each candidate's feasible terminal positions, forcing them to prioritize among accessible voter groups rather than optimizing globally. However, a longer horizon does not simply expand the feasible set—it also changes the strategic interaction between candidates. Since the value function is computed via backward induction from the terminal payoff, extending the horizon reshapes the entire sequence of mixed-strategy equilibria, potentially producing qualitatively different trajectories rather than mere extensions of shorter-horizon paths. Notably, convergence to a central position is not guaranteed even with a sufficiently long horizon; the coupling through the softmax denominator, and the nonconvex and nonconcave structure of the payoff can sustain oscillatory equilibrium behavior. The framework thus suggests that campaign duration is not merely a logistical parameter but a strategic variable that qualitatively shapes equilibrium outcomes.

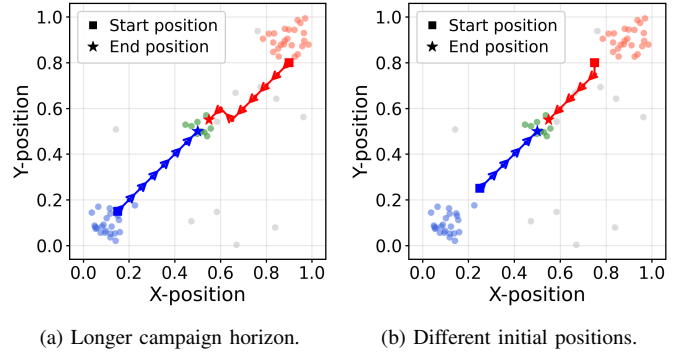


Fig. 2. Effects of horizon length and initial positions under the same polarized electorate as Fig. 1(b). The stagewise mixed-strategy Nash equilibria along the displayed paths are pure. The dots have the same interpretation described in Fig. 1(b). (a) With $T = 7$ and the same initial positions as Fig. 1(b). (b) With $T = 5$ and different initial positions $(0.25, 0.25)$ and $(0.75, 0.80)$ from Fig. 1(b).

C. Initial positions

Comparing Fig. 1(b) with Fig. 2(b), where the candidates' initial positions are shifted from $((0.15, 0.15), (0.90, 0.80))$ to $((0.25, 0.25), (0.75, 0.80))$, we observe that the initial positions affect the candidates' trajectories even when the voter distribution and time horizon remain unchanged. This path-dependence underscores that optimal strategy is also sensitive to the initial state.

V. REPOSITIONING COST

In the model of Section II, repositioning is free: a candidate may move anywhere the speed constraint allows, and only her final position affects the vote. In practice, repositioning is observed by the electorate, and a candidate who changes position often is perceived as inconsistent and loses credibility. In this section we extend the model by a cost of repositioning and study the extended game analytically and numerically.

Let $\Lambda_j(t)$ be the cumulative distance travelled by candidate j on the policy space up to time t ,

$$\Lambda_j(t) = \int_0^t \|a_j(\tau)\|_m d\tau, \quad \Lambda_j(0) = 0, \quad j = 1, 2, \quad (5)$$

where $\|\cdot\|_m$ is a *movement norm*, which need not coincide with the *speed norm* $\|\cdot\|_s$ of the constraint $\|a_j\|_s \leq c_j$; equivalently, $\dot{\Lambda}_j = \|a_j\|_m$. The observed utility of voter i from candidate j is

$$u_{ij}^*(v_i, x_j, \Lambda_j) = \lambda_{ij} - \rho_j \Lambda_j - \Phi_i(\|v_i - x_j\|_i), \quad (6)$$

where $\rho_j \geq 0$ is the credibility-loss rate of candidate j and the valence λ_{ij} of Section II is reduced by $\rho_j \Lambda_j$, uniformly across voters. The utility and the probability that voter i votes for candidate j are, as before,

$$u_{ij} = u_{ij}^*(v_i, x_j, \Lambda_j) + \varepsilon_{ij},$$

$$s_{ij} = \frac{\exp(u_{ij}^*(v_i, x_j, \Lambda_j))}{\sum_{k=1}^2 \exp(u_{ik}^*(v_i, x_k, \Lambda_k))},$$

and the terminal payoff is P_2 's expected vote count $V_2 = \sum_{i \in \mathcal{I}} s_{i2}$ at time T . Writing $\sigma(y) = 1/(1 + e^{-y})$ and $z =$

$\rho_1 \Lambda_1 - \rho_2 \Lambda_2$, the expected vote count depends on $(\mathbf{x}, \Lambda_1, \Lambda_2)$ only through $(\mathbf{x}, z)$:

$$q(\mathbf{x}, z) = \sum_{i \in \mathcal{I}} \sigma \left(u_{i2}^*(v_i, x_2, 0) - u_{i1}^*(v_i, x_1, 0) + z \right), \quad (7)$$

$$L = q(\mathbf{x}(T), z(T)).$$

The cumulative movements Λ_1 and Λ_2 augment the state of the game, but by (7) the payoff depends on them only through z with autonomous dynamics,

$$\dot{z}(0) = 0, \quad \dot{z} = \rho_1 \|a_1\|_m - \rho_2 \|a_2\|_m, \quad (8)$$

which involve $(\mathbf{x}, z)$ and the actions alone. The terminal payoff and the dynamics of $(\mathbf{x}, z)$ depend on the full augmented state only through $(\mathbf{x}, z)$. Thus $(\mathbf{x}, z)$ is a sufficient state for the Markov formulation, and we use feedback strategies $\gamma_j(t, \mathbf{x}, z)$. Since z is a function of the observed position history, the augmentation does not introduce any additional private information. The cost of repositioning is not a stage cost: $g \equiv 0$ still holds, and the cost enters only through the voters' utilities and hence through the terminal payoff. Since the electorate is fixed, the game remains constant-sum.

Assuming a classical solution exists and letting $W : [0, T] \times \mathbb{R}^{2n} \times \mathbb{R} \rightarrow \mathbb{R}$ denote the value function on the reduced state $(t, \mathbf{x}, z)$, the Bellman–Isaacs equation reads

$$-\frac{\partial W}{\partial t} = \min_{\|a_1\|_s \leq c_1} \left[\left\langle \frac{\partial W}{\partial x_1}, a_1 \right\rangle + \rho_1 \frac{\partial W}{\partial z} \|a_1\|_m \right] + \max_{\|a_2\|_s \leq c_2} \left[\left\langle \frac{\partial W}{\partial x_2}, a_2 \right\rangle - \rho_2 \frac{\partial W}{\partial z} \|a_2\|_m \right], \quad (9)$$

with terminal condition $W(T, \mathbf{x}, z) = q(\mathbf{x}, z)$. The Hamiltonian is additively separable in (a_1, a_2) , so the Isaacs condition holds. Assume that a continuously differentiable function W satisfies (9) and the terminal condition, and that admissible feedback strategies attaining the Hamiltonian extrema exist. Then the verification argument of Proposition 3.1 applies on the augmented state space $(\mathbf{x}, z)$. When the movement norm coincides with the speed norm, $\|\cdot\|_m = \|\cdot\|_s$, the extrema are given by (1) and (9) becomes

$$-\frac{\partial W}{\partial t} = -c_1 \left(\left\| \frac{\partial W}{\partial x_1} \right\|_{s,*} - \rho_1 \frac{\partial W}{\partial z} \right)_+ + c_2 \left(\left\| \frac{\partial W}{\partial x_2} \right\|_{s,*} - \rho_2 \frac{\partial W}{\partial z} \right)_+, \quad (10)$$

where $(y)_+ = \max\{y, 0\}$, attained by $\gamma_j^*(t, \mathbf{x}, z) = \mp c_j \hat{a}_j \mathbf{1}\{\|\partial W/\partial x_j\|_{s,*} > \rho_j \partial W/\partial z\}$, with $\hat{a}_j$ any maximizer of $\langle \partial W/\partial x_j, a \rangle$ over $\|a\|_s \leq 1$, the sign $-$ for P_1 and $+$ for P_2 . Compared with (4), a candidate moves at full speed along her gradient only when the dual norm of the gradient exceeds $\rho_j \partial W/\partial z$, the marginal cost of credibility, and does not move otherwise; at $\rho_1 = \rho_2 = 0$, (10) reduces to (4). Since q is increasing in z and the increments of z in (8) do not depend on z , a comparison argument gives $\partial W/\partial z \geq 0$. The same multiplier $\partial W/\partial z$ appears in both candidates' thresholds.

In the following, we explore the numerical effects of the repositioning cost. Similarly to Section IV, the stage actions

are $a_j(t) \in \{-h, 0, h\}^2$ with $h = 1/(G-1)$, so the discrete speed norm is $\|\cdot\|_\infty$. We take the movement norm to be ℓ_1 , so that the credibility loss is charged per coordinate moved:

$$\Lambda_j(t+1) = \Lambda_j(t) + \|a_j(t)\|_1, \quad \Lambda_j(0) = 0, \quad (11)$$

an axis-parallel move adds h and a diagonal move $2h$, and the reduced state evolves as $\mathbf{x}(t+1) = \mathbf{x}(t) + (a_1(t), a_2(t))$, $z(t+1) = z(t) + \rho_1 \|a_1(t)\|_1 - \rho_2 \|a_2(t)\|_1$. With this pairing the brackets in (9) separate across coordinates,

$$\min_{\|a_1\|_\infty \leq c_1} \left[\left\langle \frac{\partial W}{\partial x_1}, a_1 \right\rangle + \rho_1 \frac{\partial W}{\partial z} \|a_1\|_1 \right] = -c_1 \sum_{k=1}^n \left(\left| \frac{\partial W}{\partial x_{1,k}} \right| - \rho_1 \frac{\partial W}{\partial z} \right)_+,$$

and likewise for P_2 with the opposite sign: each candidate decides coordinate by coordinate whether the positional gain exceeds the credibility cost. The value function is computed by backward induction on the reduced state as in Section IV, the (a_1, a_2) entry of the stage matrix being $W_{t+1}(\mathbf{x} + (a_1, a_2), z + \rho_1 \|a_1\|_1 - \rho_2 \|a_2\|_1)$. At $\rho_1 = \rho_2 = 0$ the model reduces to the baseline game.

We use the discretized game of Section IV with the ℓ_∞ speed constraint of the grid and the ℓ_1 movement norm (11), the utility model of Table I with a different electorate, and a horizon of $T = 8$ stages. The electorate consists of $I = 59$ voters in three clusters: a *green* cluster of 14 voters near $(0.34, 0.46)$ and a *purple* cluster of 19 voters near $(0.37, 0.27)$, both with acceptance radius $r_i = 1.2$, and an *orange* cluster of 20 voters near $(0.73, 0.56)$ with $r_i = 0.5$, together with 6 background voters with $r_i = 0.5$. The candidates start at $x_1(0) = (0.45, 0.60)$ and $x_2(0) = (0.70, 0.70)$, so that initially the green and purple clusters favor P_1 and the orange cluster favors P_2 . We solve the game numerically for every pair of credibility-loss rates $(\rho_1, \rho_2) \in \{0, 1, \dots, 12\}^2$.¹ Fig. 3 shows the clusters, the initial positions, and the equilibrium trajectories for two of the 169 rate pairs $(\rho_1, \rho_2) = (4, 2)$ and $(\rho_1, \rho_2) = (2, 10)$.

A. Contest index

We classify, for each pair of rates (ρ_1, ρ_2) , which part of the electorate the two candidates compete over. Initially the green and purple clusters favor P_1 and the orange cluster favors P_2 ; we call the former P_1 's *home side*, with joint centroid $\bar{v}_h$ (the average of its 33 ideal points in the two home clusters), and the latter P_2 's *home side*, with centroid $\bar{v}_o$. A contest over a side is a movement of *both* candidates toward it: one candidate moves toward the opponent's side and the opponent moves back toward her home side to defend it. For each candidate P_j and each side $c \in \{h, o\}$ we measure the distance she gains on that side over the campaign,

$$d_{jc} = \|x_j(0) - \bar{v}_c\|_2 - \|x_j(T) - \bar{v}_c\|_2, \quad (12)$$

where $x_j(T)$ is the expected terminal position under the equilibrium strategies and the Euclidean distance is the

¹We solve the finite game numerically and propagate the resulting equilibrium state distribution deterministically, without trajectory sampling.

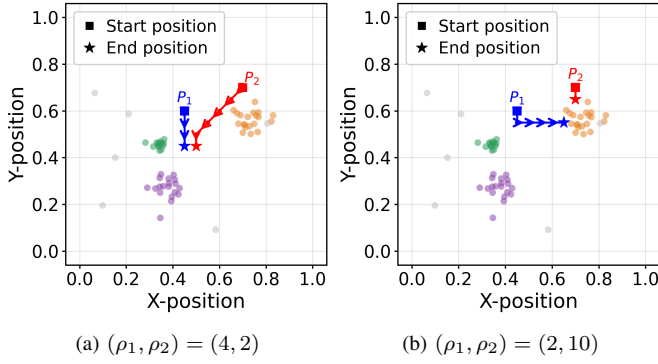


Fig. 3. Electorate and initial positions of the repositioning-cost experiments, with equilibrium trajectories for two pairs of credibility-loss rates ($T = 8$). The stagewise saddle points along both displayed paths are pure. (a) At $(\rho_1, \rho_2) = (4, 2)$, P_2 moves four diagonal steps and one axis step toward the green cluster, stopping at $(0.50, 0.45)$, and P_1 moves back toward it. (b) At $(\rho_1, \rho_2) = (2, 10)$, P_2 moves one step and stays, and P_1 moves to the edge of the orange cluster.

distance in which voters evaluate positions in (6). The joint approach of the pair to side c is $A_c = d_{1c} + d_{2c}$, and we define the *contest index*

$$F = A_o - A_h. \quad (13)$$

To read the sign as a contest we require that both candidates move toward the same side by a threshold $\tau = 0.045$, slightly less than one grid step. We say that the contest is over side c when $d_{1c} \geq \tau$ and $d_{2c} \geq \tau$. Across the 169 rate pairs the contest is over P_1 's side in 63 cells and over P_2 's side in 89; the two cases never coincide, and F has the corresponding sign (positive for contest over P_2 's side, and negative for P_1 's side) in all of these cells but one. The remaining 17 cells, located near the regime boundary or at the corners of the rate plane, represent special cases described in the caption of Fig. 4.

Overall, $F < 0$ means that the two candidates jointly gain more distance on P_1 's home side than on P_2 's, and hence contest over P_1 's side; $F > 0$ means the reverse.

B. An approximately linear transition

Fig. 4 reports F over the grid of rates. The boundary of the two regimes of F with different signs is approximately linear in the (ρ_1, ρ_2) plane. A least-squares fit of the crossings gives $\rho_2^* \approx 3.26 + 0.34\rho_1$ (dashed line). Below the line ($F < 0$), P_2 moves toward P_1 's home side, and P_1 moves back toward it. Above the line ($F > 0$), P_2 stays near the orange cluster, and P_1 moves toward it. Fig. 3 shows one equilibrium on each side of the line.

The boundary is a consequence of the interaction between the two rates and the geometry of the electorate. A move toward the opponent's side wins votes there, loses votes on the candidate's own side, and costs credibility with every voter at her own rate. For this electorate, grid, and horizon, lower ρ_2 relative to ρ_1 is generally associated with joint movement toward P_1 's side, whereas higher relative ρ_2 is generally associated with movement toward P_2 's side. The fitted line summarizes these numerical results and is not asserted to be a general comparative-static boundary.

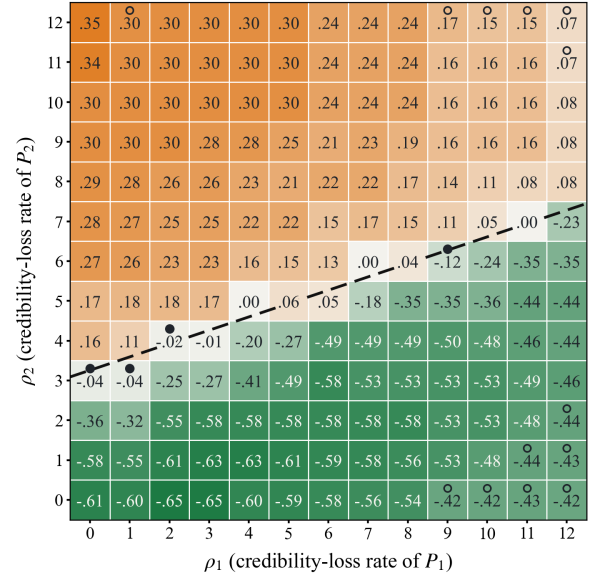


Fig. 4. Contest index F of (13) over the credibility-loss rates (ρ_1, ρ_2) with $T = 8$, ℓ_∞ speed constraint, ℓ_1 movement norm. Negative (green) cells indicate greater joint movement toward P_1 's home side; positive (orange) cells indicate greater joint movement toward P_2 's. Dashed line: least-squares fit $\rho_2^* \approx 3.26 + 0.34\rho_1$ of the sign boundary. Filled dots: the four cells in which the candidates move toward opposite sides; open circles: the 13 cells in which only one candidate moves by at least τ .

VI. CONCLUSION

In this paper, we propose a constant-sum dynamic game model for electoral competition. This framework captures both the strategic complexity of candidate behavior in a stochastic environment and the simultaneity of campaign decisions under complete information. Numerical examples are used to illustrate the implications of the model.

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